

Rock Hardness: Mark or Groove?

BUILD • Science • 40 minutes

I can explain relative hardness and identify a fair, safe scratch method for comparing rock samples.

Prepare before pupils arrive

Setup: Pretest reputable educational samples and the secured scratch jig. Prepare fixed guide length/passes, separate trays, after-wipe macro cards and a complete no-tool data route.

Safety: The local risk assessment leads. Teacher operation is a complete route. Remove unstable samples, use eye protection where assessed, contain any debris and stop rather than forcing a clearer result.

Equipment: Two teacher-pretested coded dry rock samples; stable trays; secured blunt scratch jig or teacher demonstration; fixed guide; wipes; hand lens; result card; eye protection where required by local assessment.

Safety: Follow the school's current rock-testing risk assessment. No unknown, sharp, dusty or crumbling samples; no hammering, breaking, tasting, smelling or dust blowing. An adult controls the secured tool. Stop immediately on flaking or dust and wash hands.

Fallback: Direct the adult tester, use macro before/after cards or interpret the fixed data. Handling the rock or tool is never required.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Rocks: test hardness.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 4 min	Explain why the W9 sorting question stayed unchanged. Name one visible rock property and one property that needs a test.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter · 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 5 min	hardness — resistance to a scratch groove — a line cut into the surface control — what we keep the same Place a teacher-prettested dry sample in a stable tray. Use only the secured blunt scratch jig or a teacher demonstration.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 5 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	Compare outcomes with cautious words: deep groove, shallow groove, faint groove or no groove under this method. A clearer remaining groove means lower resistance to this particular scratch method; it does not measure every kind of strength.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Place each method or observation under the role it performs in a valid hardness comparison. Lock the stated method.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 10 min	Plan and direct a safe two-sample relative hardness comparison, then record the method, groove observation and bounded conclusion. Record: A Rock Hardness Method Card with prediction, changed sample, controls, after-wipe observation and a relative conclusion within the stated method.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Explain how the method distinguishes a tool streak from a groove and name two controls. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move; written closure is optional when Voice and Audience are observable.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: No groove remained; this sample resisted the stated scratch method.

Yes. Under this stated method, no groove remained after wiping. That is a relative scratch-resistance observation.

Misconception repair

Repaired. The scratch method compares relative hardness only; it does not test weight, toughness or construction strength.

Guided checkpoint

Yes. Under this stated method, no groove remained after wiping. That is a relative scratch-resistance observation. Record only what remained after wiping, and keep the claim within this method.

Deep indentation remains

VALID GROOVE EVIDENCE. Valid evidence: the sample showed low resistance to this stated scratch method. The deciding observation is the indentation that remains after wiping.

Shallow indentation remains

VALID GROOVE EVIDENCE. Valid evidence: a shallow groove remained under the method. Do not discard a faint or shallow remaining indentation.

Line wipes away

VALID NO-GROOVE EVIDENCE. Valid evidence: the sample resisted a groove under this method. Check the surface after wiping, not the first pale line.

Same tool, guide and passes

KEEP THIS THE SAME. Control: the relevant method is kept as consistent as practical. This card describes fairness, not the result.

The sample feels heavy

DOES NOT MEASURE HARDNESS. Not hardness evidence: mass or weight does not answer the scratch question. Match the observation to resistance to scratching.

The edge flakes or dust appears

STOP / USE PARITY ROUTE. Stop. Preserve the note and move to a safe pretested sample or representative parity data. Safety is part of the method; do not press harder or continue with dust.

Model limitation

This is a relative classroom comparison with one tool and method. It does not give a Mohs number, prove construction strength or represent every specimen of a named rock.

Independent evidence

A Rock Hardness Method Card with prediction, changed sample, controls, after-wipe observation and a relative conclusion within the stated method.

Supported: Choose groove/no-groove after-wipe cards for two samples and direct the adult through the same method. Use pictures for tool, passes, wipe and groove. Complete: After wiping, __ remained.

Standard: Compare two coded samples with the same stated method and justify which resisted a groove more. Use: We changed __. We kept __. Sample __ showed __, so under this method __.

Stretch: Evaluate the controls, distinguish hardness from strength and propose one safe repeat or method improvement. Preserve the first result and explain why the improvement would make the comparison more dependable.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: [SCI_B_W10A_Rock_Hardness_Explore.html](#)

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.